

Phase 5 – Project Development Phase

Project Title

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Introduction

The Project Development phase focuses on implementing the planned features in the ServiceNow platform. During this phase, users, roles, custom tables, fields, and Access Control Lists (ACLs) were created and configured to enforce secure record-level access based on the user's role and the Branch field.

2. Development Environment

Component	Details
Platform	ServiceNow Developer Instance
Language	JavaScript (ACL Script)
Security	Access Control Lists (ACLs)
Database	ServiceNow Custom Table
Browser	Google Chrome

3. User Creation

A test user named EEE User was created with the following details.

Field	Value
User ID	EEE User
First Name	EEE
Last Name	User
Email	eeeuser@gmail.com

4. Role Creation

Four custom roles were created.

Role	Purpose
bb1	Read Access
bb2	Create Access
bb3	Write/Edit Access
bb4	Delete Access

All four roles were assigned to the EEE User.

5. Custom Table Development

A custom table named Institution Details (u_institution_details) was created.

Table Fields

Field Name	Data Type
Student Roll Number	Auto Number
Student Name	Reference (User)
Faculty Name	Reference (User)
Branch	Choice
Email	String
Phone Number	String
Description	Multi-line String

dev395038.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D2ae8930c3630f10949353f1b4013106%26sysparm_record_target%3Dsys_db_...

servicenow

Table - Institution Details

Search

Table Institution Details

Delete Update Delete All Records

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label

Application

* Name

Remote Table

Columns Controls Application Access

Table Columns for text Search 1 to 13 of 13 New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Student Roll Number	String	(empty)	40		false
Faculty Name	Reference	User	32		false

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dev395038.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D2ae8930c3630f10949353f1b4013106%26sysparm_record_target%3Dsys_db_...

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Table - Institution Details

Search

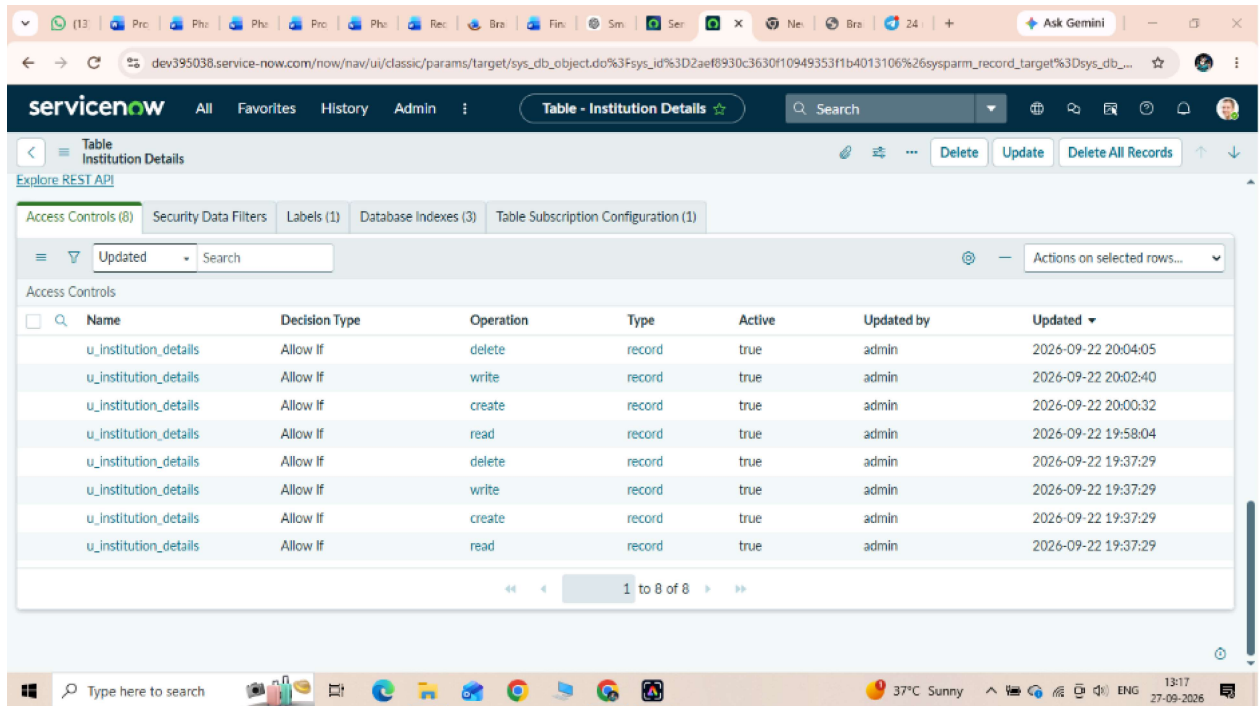
Table Institution Details

Delete Update Delete All Records

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Student Roll Number	String	(empty)	40		false
Faculty Name	Reference	User	32		false
Updated	Date/Time	(empty)	40		false
Student Name	Reference	User	32		false
Branch	Choice	(empty)	40		false
Created by	String	(empty)	40		false
Email	String	(empty)	40		false
Description	String	(empty)	40		false
Phone Number	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Insert a new row...					

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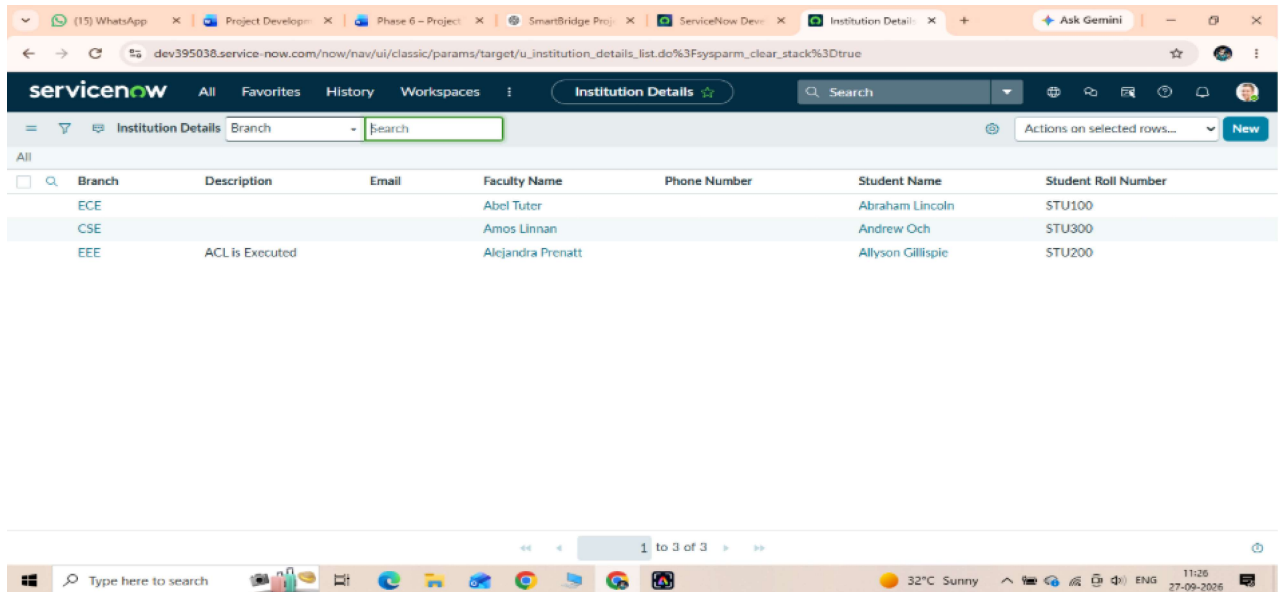


6. Sample Record Creation

Multiple student records were created using different branch values.

Student	Branch
Student 1	EEE
Student 2	ECE
Student 3	CSE

These records were used for testing ACL behavior.



7. ACL Implementation

Read ACL

Property	Value
Type	Record
Operation	Read
Table	u_institution_details
Role	bb1
Condition	Branch = EEE

Script Used

JavaScript

```
(function () {  
    if (gs.hasRole('admin')) {  
        return true;  
    }  
  
    if (gs.hasRole('bb1')) {  
        return true;  
    }  
  
    return false;  
})();
```

The screenshot displays the ServiceNow interface for configuring an Access Control rule. The rule is named 'u_institution_details' and is set to 'read' operation. The decision type is 'Allow If'. The description states: 'Allow read for records in u_institution_details, for users with role bb1, and if the ACL script returns true, and if the ACL condition (u_branch=EEE^EQ) evaluates to true.'

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Script

Turn on ECMAScript 2021 (ES12) mode

```
1 (function () {  
2     // Allow admin users full access  
3     if (gs.hasRole('admin')) {  
4         return true;  
5     }  
6     // Allow only EEE branch users to see EEE records  
7     if (gs.hasRole('bb1')) {  
8         return true;  
9     }  
10    // Deny access for all others  
11    return false;  
12 })();  
13
```

Related Links

- [Show ACL Execution Plan](#)
- [Run Point Scan](#)

Create ACL

- Role: bb2
- Operation: Create

Write ACL

- Role: bb3
- Operation: Write

Delete ACL

- Role: bb4
- Operation: Delete

8. Development Workflow

The implementation followed these steps:

1. Created the EEE User.
2. Created four custom roles.
3. Assigned roles to the user.
4. Created the Institution Details table.
5. Added all required fields.
6. Inserted sample records.
7. Configured Read ACL with JavaScript.
8. Created Create, Write, and Delete ACLs.
9. Verified permissions using impersonation.

9. Development Challenges

Challenge	Solution
Role assignment	Verified assigned roles
ACL configuration	Tested using impersonation

Record visibility	Checked Branch condition
Permission errors	Reviewed ACL settings

10. Development Outcome

The development phase successfully implemented a secure ServiceNow application where administrators retained full access, users with bb1 could view EEE records, bb2 enabled record creation, bb3 allowed editing, and bb4 permitted deletion. The application achieved role-based record security through Script-Controlled ACLs.